The Habitable Worlds Observatory with Evgenya Shkolnik

Podcast: Startalk Podcast

AI Executive Summary

1. Core Thesis The episode argues that defining a habitable world requires moving beyond the simplified "Goldilocks Zone" conceptâ€”which considers only liquid water potentialâ€”to a complex analysis of atmospheric stability, stellar radiation, and magnetic protection. Leveraging statistical data indicating that nearly every star hosts at least one planet (6,000+ confirmed), NASA's Habitable Worlds Observatory (HWO) aims to identify biosignatures within 50 parsecs (~150 light-years). Smaller missions like the SPARKS CubeSat serve as cost-effective precursors to study ultraviolet impacts on exoplanetary atmospheres, with funding prioritized through the peer-reviewed Decadal Survey process.

2. Key Arguments & Data

3. **Exoplanet Statistics:** There is a transition from zero confirmed exoplanets in the 1990s to over 6,000 known confirmed planets today; statistically, it is reasonable certainty that every star has at least one planet.
4. **Definition of Habitability:** Distinction between "habitable" (potential for liquid water via atmosphere and stable star) and "inhabited" (presence of actual life). Earth-centric biases in defining habitability must be acknowledged despite extreme environments on Earth thriving.
5. **Star Types & Activity:** 75% of galactic stars are low-mass M-dwarfs (Red Dwarfs), dominating the planet population, yet they are more magnetically active and turbulent than Sun-like stars, creating UV hazards that can strip atmospheres or cause false biosignature detections.
6. **Mission Specifications:**
    - **HWO (Habitable Worlds Observatory):** Target location at Sun-Earth L2 Lagrange point; main mirror size between 6 to 8 meters; primary goal is a community-led survey for life on rocky planets within ~150 light-years.

    - **SPARKS:** Small UV CubeSat (9cm diameter), estimated cost $10 million, designed to characterize ultraviolet radiation from stars hosting exoplanets.
7. **Funding & Process:** HWO is a flagship project funded in the "few billions" range, selected via the Astro 2020 Decadal Survey; large-scale funding requires collaboration across nations and institutions to be justified.

8. **Search Strategy:** Search targets must balance known exoplanets with serendipitous discovery (deep field surveys), requiring a modular design to allow upgrades over time similar to Hubble.

9. Actionable Takeaways

10. **Understand Funding Mechanisms:** Recognize that major space missions like HWO are not just engineering feats but result from rigorous peer-reviewed community consensus (Decadal Surveys) which dictates funding priorities and allocation.
11. **Engage with Small-Scale Innovation:** Support or follow the progress of CubeSats like SPARKS, as they prove the cost-effectiveness of small satellite astronomy ($10M vs billions) for foundational data collection on stellar UV environments.
12. **Refine Habitable World Expectations:** Adjust expectations from "just right temperature" to a holistic model requiring stable atmospheres, protective magnetic fields, and manageable stellar UV output when evaluating potential life elsewhere.
13. **Monitor the Decadal Surveys:** Stay informed about the 10-year scientific surveys (e.g., Astro 2020), which serve as the authoritative source for future funding priorities and telescope capabilities in astrophysics.
14. **Evaluate Telescope Time Allocation:** Acknowledge that competition for flagship time is intense, but that peer review ensures the most compelling science questions receive opportunities to utilize the HWO infrastructure once operational.

Full Transcript

Matt, it was all about habitable worlds. I want to visit them. I want to, although they don't sound that habitable. Let's find out more coming right up. Welcome to StarTalk, your place in the universe where science and pop culture collide. StarTalk begins right now. This is StarTalk. Neil deGrasse Tyson, your personal astrophysicist. Today my co-host is Matt Kirshen. Matt, welcome back. Hey, thank you. It's nice to be back in town. You're based in L.A.? I am. You've just realized that. I didn't realize. I think you thought I was here all along. Oh, I'm cloned or something, but I'm delighted you're coming through town. We snatched you off the comedy line, and here you are. Yeah, delighted to be snatched. All right. All right. So you know what we're going to talk about today. Habitable worlds. Love it. Ooh. I feel like this is always a popular subject with business. Always. Habitable worlds, not Earth.

Right. Earth is like prime habitable world, but we're going to see if there's more. Nice. We have one of the world's experts on that subject, Evgenius Konik. Very nice. Did I say that correctly? You did. I did. Pretty much. Approximately said it perfectly. Welcome, you're a professor of astrophysics at Arizona State University. Indeed. and the Department of Earth and Space Science. School of Earth and Space Exploration. Oh, that's a little extra action there. That's how we roll over there. Yeah, I know. That feels like it comes with an extra costume. Like, what's the exploration? Helmets, the very least a hat. For you, you can visit us. We'll get you a hat. Nice. Distinguished Visiting Scientist at JPL. We all love JPL, the Jet Propulsion Labs. That's your hood. I've got to visit JPL. I did that like a year or two ago, and it was fantastic. Yeah, yeah, no, it's great. So great. It's great. Co-chair of NASA's Habitable World Observatory Community Science and Instrument Team that has an acronym, CSIT. CSIT, we call it. CSIT. Yes. Okay, no one else will call it that. You realize this? Okay.

And after all of that, you're also an associate up at the Harvard Department of Astronomy, Cambridge, Massachusetts. There's apparently your PI, principal investigator of SPARKS, Space Telescope Mission. What is SPARKS? SPARKS is an acronym for the Star Planet Activity Research

CubeSat, which is a small space telescope. That sounds tortured, like you all had a five hour meeting. It is not tortured. If somebody came up with this. I know about CubeSats. They're this big, very affordable, very university-friendly satellites that can be launched relatively cheaply. And you clad it with instruments, right? So what are the instruments on your CubeSat? Sparks has an ultraviolet space telescope inside of it. So it's a small telescope, as you said, that fits into a small box. It's only nine centimeters across, but it is equipped with... There's a marker. Show us what nine centimeters is. Oh dear, almost four inches. Okay, so nine centimeters. Yeah, about like that. It's adorable. It's a little adorable telescope. A little adorable telescope. Don't call telescopes adorable. That is so insulting. You could put it into your backpack. Right. Okay. The box... How much would that...

That cost yours, Jude, if I were to do that. Well, I charge you at least 10 million for it. Okay, noted. Okay, so what are you gonna do with the UV? So the ultraviolet is an important wavelength range that we need in order to study the stars that host exoplanets. So these are planets around other stars. And as we all know, the ultraviolet is a dangerous, highly energetic wavelength range, right? And so very... We are very grateful that we have an atmosphere here on Earth that protects us from the sun's ultraviolet. But there's a lot of great science that you can do in the ultraviolet. And so we wanted to launch a space telescope in order to study this UV from other stars and how they impact their own planets. So your list already contains stars with known planets. You're not looking for planets. That's right. We're not looking for planets. And most of our targets have known planets.

Not all of them have known planets yet, but statistically, they should all have planets. Oh, that's bold. If you don't have a planet, you should. We just haven't discovered it, but it should be there. Well, this is something I've learned just from actually being on your show as the host. It seems like well within my lifetime. In fact, even the last third of my lifetime, we've gone from basically assuming there are almost no works of planets to assuming just about every single star has them, right? Assuming is an interesting word there. Okay, good. Pull me up, please. She means it's a wrong word. She should be polite. No, no, we can assume it, but we also know it. This is an interesting thing you've done with your homework here. Yeah, very

interesting. This is a good professor at play. It's an interesting way of writing. See, she is a professor. She tried to be nice to her students. In our lifetimes, yours and mine, mine, we went from knowing, for sure, of zero exoplanets to over 6,000 known, confirmed...

for sure are there. Now, if you can work out the probabilities and the statistics of the known sample compared to all the ones that we have not been able to detect planets around or have not looked yet, we can say with reasonable certainty that on average, every star has at least one planet. And so just to distinguish, just to go back to her correcting you, okay? It's not that we assumed no stars had planets. We just knew that. We had no data on any stars having planets yet. Cool. Right. That's different. That is different. Not knowing is not the same as presuming. Presuming zero. Zero. Right. Did I clear them up on that? Yes. Good. Okay. So let's understand the word habitable. And I've never liked the word. I thought it was too many syllables crammed together in a row. And so. And also just like as a torrent comedian who stays in hotels.

most of my life, like I know that the word habitable has a range of meanings. Habitable hotels. So habitable, I've rejected it for quite some time. Of course, I use it if I have to, but I just prefer Goldilocks zone, Goldilocks planet, which is right in the middle of extremes. So once you formally define for us habitable. Well, if you use Goldilocks and you think about the Goldilocks zone, right, that is a very No offense. Simplistic expression of habitability because right now we say that that's just a place, a zone, a ring around a particular star where it has the right temperature. Not too close. Not too close, so you're not too hot. Not too far. Not too far, so you're not too cold. So it's just right in our sense, meaning that you could have liquid water on the surface of it. But what else do you need, right? You need...

the water to be there, in order for the water to be there, you need the atmosphere, you need to- It's not good enough just to be able to have liquid water, you gotta have the liquid water. That's right. Exactly. Like water would be liquid if it existed in this condition. If it existed. That's right. That's right. If water were on this planet and it had an atmosphere and it had a well-behaved star, then that planet has the potential to be habitable. that

doesn't mean that it is inhabited, which is a completely different thing that we have to think about. But all the pieces need to fit together to be habitable. So habitable is different from habitated. Yes. That word doesn't even exist. But we all know what I meant. It's your show. I get to use whatever word I want. So what about our inherent bias as earth life?

How biased are we? Is that habitable? Is it just like Earth? I mean, I remember, I don't remember. I read that in Darwin's day, there was this idea, if life evolved, as he described, you might need a tide pool that's room temperature, and you start constructing these things that are just fine for us, or what might be our genetic ancestors. Meanwhile, we're finding life form thriving in extreme environments on Earth. So this whole thing about the tepid environment in which to evolve. I know there are like these extrema files on earth that can exist in like near volcano. Do the backstroke in a volcano? Well, not in. No, they're volcanic vents. Right. It's still a liquid water state, but they're getting their energy from other places. Ludicrously hot. Or ludicrously hot. Or ludicrously acidic or alkaline. Exactly. So my question is, how biased are you? Very.

Okay, that's it. But of course, we have to start with what we know. So we know, we understand how surface life on the Earth interacts with our atmosphere and creates these biosignatures. These are chemical fingerprints of surface biology. That is the kind of thing that we're gonna be looking for in other planetary systems because we can only right now look for these chemical signatures in planets' atmospheres. And so therefore, why not look for things that we know and we understand? And if you think that we are only in the life as we know it crowd, really, it's possible that life is very similar to us, it's possible that it's very different, but looking for what we know is critical. And then, another thought that I had with regards to the detection of life is that in astronomy, for instance, we have never detected just one of anything, right? So we have a first detection of a pulsar, and then we learn how to find more pulsars, right? We get better at it, more efficient at it, and through a statistical sample of pulsars, we learn something about pulsars. And this goes way back. We thought the sun was something special and unique. We learned that the other dots of light in the sky are just other suns. Right. And then we thought, Earth is surely, no, we're just another.

Object orbiting the Sun so it's kind of the attack on human ego Is that this scientific exercise in the history of astronomy and so but I'd like that press given that precedent you're just pushing it forward exactly given that we have Detected life on Earth, which I think we can agree with here on earth here in this room Then there should be other life on other planets, and it's possible that there's life very similar to ours at least chemically around all these other stars All right, so how do you bring the biology into the questioning and the experiment design? How much biology do you formally have? Well, you just reach over to the biology department and grab somebody. That's what I would do. Get someone where that's their expertise. Absolutely. And say, what should we be looking for? Absolutely. I work with people who are astrobiologists.

These are astronomy people, focused people. There's biology people and they come together and they call themselves as a group, astrobiologists. But yes, people have to have a deep expertise in some field and then be able to communicate with people across fields. And we have these wonderful interdisciplinary studies in order to work out what is habitable and whether or not we can discover if it's inhabited. Might their deep expertise itself be a bias in the sense that if they're an expert in all life on Earth, and they celebrate the biodiversity on Earth. At the end of the day, all life on Earth has common DNA. Are they capable of thinking about life that has no DNA at all? See, what I'm hearing from this as well is that I'm at a big advantage in this whole conversation by having no expertise.

I'm coming to this with zero bias compared to you scientists. So it's easier for you to say, how about life with no DNA? Yeah, exactly. I'm like, I can throw all of this out and I'm probably, you can all share in my Nobel. Well, that's exactly what we need. We need diverse opinions. Right. And that's what we have right here. Yeah, we have all the experts and then one guy who just winging it. and asking all the hard questions, right? These are the folks who are going to ask us questions. The hard questions like, I just got stuck in the door again. Can someone free me? That's all these important. That's right. And what is it about human evolution that causes such a situation? Getting stuck in doors. Such a good teacher. Always seeing the good side. Space fact. Dark matter is 85% of all the gravity we measure in the universe. And we don't know what's causing it.

It doesn't interact with our light. It doesn't even interact with itself. Which means it cannot coalesce to form solid objects, like stars or planets or people. If you like that fact, you can find 4,999 more in Lost in Space. 5,000 facts. to help navigate the universe. Lost in Space is now available wherever you get your books. So the Habitable World's Observatory, is that a single telescope or multiple telescopes or a consortium? What is that? The Habitable World's Observatory, which we call HWO for short, is a single telescope, one observatory, but with multiple instruments. So it can do lots of different science in different wavelengths at different resolutions. but all the instruments are fed by one mirror. Got it. Very, very efficient and convenient. It's technically a series of mirrors, but it's one telescope. So it's a mosaic of mirrors. It will be. It's not built yet. No. The habitable world's observed to worry. But when are we going to have it? Well, we're working to do it as fast as we can, but it is NASA's next big flagship, so it's a giant mission.

following on the Hubble Space Telescopes, the James Webb Space Telescope, the Roman Telescope, and now the next one in line is the Habitable Worlds Observatory. So I was on the year 2000 Decadal Survey of astronomy and astrophysics, but every 10 years, my people get together. Okay. And we say, the most trusted among us, and we say, what are we going to do next? And if a chunk of money lands either from the NSF or from the... feds in any other department, or even private money. It helps us channel the money for the single big projects that no one institution could probably fund realistically. So what's the total cost of the Habitual Worlds Observatory? Well, that is unclear until we figure out what exactly we're going to be building, which instruments we're going to have, which countries are going to become our partners, right? So these missions are so big that Lots of countries, everyone in the world wants a piece of it and wants to contribute. So it's going to be a lot in the few billions range. Do we know where it's going to sit? The goal right now is to go to L2, which is Lagrange point two. Space-based. So that sounds like it would be the Earth-Sun L2 because then it's opposite the Sun from Earth. That's right.

beyond the orbit of the moon. The moon still goes around Earth, but it's way out there. Exactly. Right. And I think JWST is there as well. JWST is there, and Roman is going to be there. Oh, it's a whole family. Yes. It's a

very nice, stable, chill environment. Isn't that beautiful? I feel like Hubble got lonely. I know. Every so often there's like an astronaut who comes and visits. Except Hubble got service five times. Right. These... unless we AI a robot and send it out there. No one's going to get out there. We are hoping to have HWO serviceable, robotically serviceable. Wow. That is in the plans right now. We are working towards making that happen because we want to be able to upgrade it just like we did the Hubble Space Telescope, that you could take an old, like after say a few years, you want to upgrade with latest technology and instrument. The goal is to make it modular so you could have a robot come in, grab the old instrument.

Plug in the new one. It would be the first modular telescope at L2. That is correct. Nice. Very cool. Nice. I know. Yeah, because these projects get on so far in life. You're sort of limited by not just even the technology now, but the technology several years ago when it was all being initially developed. That's right. So the opportunity to upgrade it with time is not something we're going to pass up. And I just want to distinguish the Sun, Earth, L2. from the Earth-Moon L2, right? I wasn't going to pull you out, but yeah. You were thinking this, too. Yeah, we were all thinking it. So, you have the Sun, the Earth, and this Lagrangian point out there. Earth and that Lagrangian point orbit together around the Sun, okay? So it's stable out there. But the Earth-Moon L2 would have to be always on the other side of the Moon as the Moon orbits Earth.

So right because every two-body system has these five Lagrangian points and so you have to specify Which those are that's right. So how big is the mirror? The main the main the main mirror is going to be somewhere between six and eight meters. Okay, so at that size What launches it? That's a that's a good question and that's actually part of our limitation is what launch vehicles are going to be available to us say in a few years time. And big enough. And so it has to be big enough and be able to carry the weight of the whole thing. And you need some clever engineers to fold it up somehow. Exactly, then the plan is into the faring. Exactly, right. So it's very similar to how the James Webb Space Telescope launched in 2021.

And the size of the mirror is really driven by the science, right? So there's some really incredible science that we want to do and it requires lots of

sensitivity and the bigger the mirror. In my field, the bigger, the better. Okay. In some cases because we talked about sparks earlier. Yeah. And it is very small. It's not calling it adorable. It is adorable. Don't call scientific instruments adorable. I'm not a offended man anyway. It's an adorable. Small but mighty. Small but mighty, that's right. And so even though there are some science cases that really demand a large mirror, like a six to eight meter mirror, there are some science cases that could get away with a smaller mirror, but really have more time on the target or more agility or being able to work together at multiple wavelengths. I mean, there are science cases that bigger is not always better.

Yeah, so agility is the word there. Perhaps, yeah. Will the habitable world's observatory target already known planets, or does it have a search mode of its own? Well, for the driving science case of looking for Earth-like planets around sun-like stars for life on them, it's going to be searching for them. We will not know necessarily ahead of time that these planets are there, so it's going to find the planets and then go back and characterize the planets. So the telescope is one stop shopping? Yeah. Wow. You create your own data set that you then investigate? 100%. Wow. That's right. So it's going to be a finder scope and a life finder. Okay. But are you going to at least target the 6,000 or more exoplanets? Yes, absolutely. That's free data right there. You just go on in. Well.

It's not free. The operations caused will matter, and the competition to get time on this telescope will be fierce. So you're going to have to have a really great science experiment, a great idea. Explain to everybody what that means, competition for a telescope. I don't think people think that way, the way we do about that. Sure. Astronomers cannot build necessarily everyone their own space telescope, even though we at ASU have sparks. But it's a small telescope so you could probably get into space just on an institution level. But the big ones are so big that we all have to work together and get multiple governments working together and therefore it's available to anyone to apply for time.

including you, Matt. Anyone can. You don't have to be a professional astronomer. You don't need to tell me twice. Yeah, you can apply for time. Anyone can put in a proposal. And then those proposals and the thousands

of them come in, go to a review panel of experts and they decide. Peer review. Peer review, exactly. Wow. Competitor review sometimes. But it's a peer review process where the best science wins. And then, so for instance, for the Hubble Space Telescope, there's about a thousand different experiments that it runs a year. probably comparable to the James Webb Space Telescope. And so then they divide up the year's time on the telescope amongst a bunch of different experiments. Confirm for me, the HWO came out of one of these decadal surveys. Which one was it? It came out of the Astro 2020. 2020? Yes, the 2020 decadal survey, which I got to sit on and got to be part of the committee that recommended. That sounds suspicious. No, it's just part of it. It was long before I was involved, but I have long been believer in needing something like this to do the most compelling science. And the panel has, what, two dozen people on it? How many people? Well, there's the steering committee that has about that. And then there's a bunch of subpanels that recommends different kinds of things. So the panel that I was on was one that evaluated space telescopes of a particular wavelength. And then there was another panel that evaluated ground-based telescopes and another one that evaluated different kinds of activities.

I'm a people. No, it's a great process. It's a great process because what it's doing is it's bringing together the community's priorities and funneling it through a review process that then can be documented and given over to agencies and saying, these are the most compelling questions. These are the questions that demand answers. And even private funders if they want to get in on the action. But for something like that, you don't need to bring sparks to the table because it's not costly enough. Is that correct? Yes and no. Sparks as an individual space telescope does not need to go through the process because it's far too small. But we did... And total cost? About 10 million. And the total cost of having a world of... Is a few billion. Yes, it's in the noise of a giant telescope 100%. Which, you know, if you ask me and really anyone who has built a small satellite is that it totally punches above its weight.

Because it's important to study the stars. In Sparks case, what we're doing is we're studying the ultraviolet from the stars that host exoplanets to understand the context in which all these biosignatures are going to be

evaluated. But if ultraviolet is hostile to life, last I checked, because it has energies that will bust out, bust a move on your DNA, why is that a useful wavelength for you in the search for life? Well, it's only damaging if you have so many UV photons that you're chemically modifying the atmosphere or eroding the atmosphere completely such that you don't have an atmosphere anymore and therefore no liquid water. So, but we want to understand all this, right? But it can also, even if you have an atmosphere, it can chemically modify the atmosphere such that it can indicate that there is life when there isn't any or destroy biosignatures when...

when there is life. So there's an opportunity for both a false positive and a false negative detection if you don't understand the UV of star. So you want to characterize that part of the spectrum. Exactly. A question. This specializes in low mass stars? Sparks? Sparks does specialize in low mass stars. But if it's a low mass star, where are you getting your UV from? The star still admits in the UV is still enough for you to care about. Yes. Oh. Enough for us to care about. Because low mass stars are very, they're not very hot. Okay. In fact, they're quite cool. In fact, they're even called cool stars. Okay. Even though you would still vaporize in it. Right. And they're sometimes called red dwarfs. Yeah. Yeah. Red dwarfs. Yeah. So, so.

Okay, so you're looking at the little bit of UV that they still emit. That's right. And it is true. We call them red dwarfs because there are literally a million more red photons than there are UV photons. So that is true. But the UV, as I said, not every wavelength is equal. The UV is super important. And so developing a space telescope just to study the UV is very valuable. Okay. Now, also these, because I'm a lot of my... professional research was in stars and stellar evolution. So I remember that these low mass stars are highly turbulent in their atmosphere. They're busting a move every time you look away and look back. You hear about solar flares here every now and then, you know, coronal mass ejections, all right? They'll show up in the news every few months. That'd be an hourly weather report.

around these stars, wouldn't it? That's a surprise because again, from my complete layperson, I would have thought the more energetic stars would have a lot more of that going on. I know. So get us out of that. Sure. It is true. What you say that these low mass stars are far more active. They have

stronger magnetic fields at their surfaces. They can create more flares, more energetic flares, potentially more coronal mass ejections. We actually don't know because we've never really confirmed a coronal mass ejection on any of these stars before. But we expect it. We expect it. And just to be clear, the star is a plasma. So it's got free electrons that can lock into a magnetic field. So if the magnetic field does things, so does the gas with it. So if the magnetic field pops off the surface, it takes material with it. Yeah, they're locked together. It's a maelstrom. It's an awesome maelstrom. OK. So now you want to have stable life around a planet with an unstable star.

That's a tall order. Well, we don't know. We don't know how unstable those environments really are until we go and look and characterize them. And we also... That's the answer. I'm trying to say, theoretically, that's your net. We don't know. We're going to look. That is the scientist's answer. We can assume or we can go check. But there are many reasons to focus on the low mass stars initially. I mean, of course, we want to look for planets around sunlight stars because we... know that a habitable planet can exist around a sun-like star. How do we know that? I feel like I should get this one right. Look, I'm looking around the room. We can't be in the clues, is that? Yeah, we have at least one data point on that. Yeah. That's right. So we know, we know that that's possible. We don't know if the answer around the M star is yet, but the reason why It's important to understand this is because 75% of all the stars in our galaxy are these low mass stars. They are the dominant hosts of habitable, potentially habitable planets. So most of the habitable or potentially habitable real estate is in fact around M stars in our galaxies. And so that's the coolest category of stars. Yes. And the letter means absolutely nothing. Okay. Don't try to give meaning to it. All right. M stars and that there's a sequence of letters that has historical accident to it, O B A F G K M in downward sequence of temperature. So these M stars dominate the galaxy and therefore their planets dominate the planet population. And so, like me, even though I haven't been working on Habitable Worlds Observatory for two decades because that hasn't existed, I have been thinking about habitable planets and I've been guided by one giant science question my entire career and that is, what is the distribution of habitable environments in our galaxy? And in order to answer that, understanding planets around these low mass stars is a huge part of the question. Now you said that very broadly, habitable environments in the galaxy. So that blows

open the Goldilocks concept into I presume you can have a habitable region in a moon around a planet. That's right. For example, right? It doesn't require a distance from the sun if you have a heat energy source from somewhere else. So you're open to anything and anywhere that could happen. Well, I'm open to it because I want to understand it. Now, different experiments have to focus on different regimes. So for instance, the Habital World Observatory is looking at...

Surface life as opposed to subsurface life, right? There's an excitement at subsurface life. That's right. Therefore, we're not working about it. Exactly. So I'm curious. I'm certainly interested. I thought I had a gotcha question. No, of course not. Not with HWO. That's right. With not with HWO, I mean, we could potentially think about, you know, in our own solar system, right, studying the plumes of Enceladus as an example. But we're not going to do that with HWO. Enceladus, do you mind me, that's Jupiter or Saturn? Saturn, it's a bit of Saturn. And that's under those are plumes that come from beneath the surface. That's right, that's right. So there's some speculation that there might be conditions for life there. But we're not going to study those on moons in other solar systems. So tell me about a magnetic field.

and planets. Do you guys think about the need for those? The Earth has a magnetic field, and that we think that it protects our atmosphere from ejections from the Sun, like you said earlier, coronal mass ejections, and that... Because they're charged particles. Because they're charged particles. We're also getting charged particles from the rest of the galaxy. It's not just from the Sun, but primarily from the Sun. And so we think that the Earth's magnetic field protects the Earth's atmosphere. But it's not necessarily true, or we don't know for sure, if that is a requirement of a habitable environment. So it depends. What is the star doing? What, how thick the atmosphere is? What, what are the conditions on the surface? So you worry that if we didn't have the magnetic field, then these High energy charged particles might be hostile to life on the surface. That's right. Got it. That's right. It would strip in a few ways. It could strip away our atmosphere. It could chemically modify our atmosphere. And it could even the strongest ones would penetrate down to the surface and interact with life on the surface. And that could be very damaging as well. What she said in Iraq,

she means kill. Okay. I just want to make that clear. Got ya. Damage, you know.

So we are interested in magnetic fields, and so we have so far really have only indications of magnetic fields on Jupiter-sized planets outside of our solar system. We do not yet have any measurements of magnetic fields on small earth-sized planets outside of our solar system. So this is also another bit of research that I'm very excited about in the future. Are there other CubeSats that will be in tandem with this current one here at Sparks? Right now, no, you got it right. Sparks is launched and it's already up and running and doing its job. In the future, what we want to do is potentially put together hundreds of these small satellites, say in the radio wavelength range. We have this concept called the mega wave radio surveyor, where one of the driving science cases might be to look for Earth.

Size magnetic fields around Earth's inhabitable zones. So this is something that is even beyond the Habitual Worlds Observatory. They really want to listen for TV on other radio waves. They want to look for it, but she's not letting on to that. I mean, I'm over TV credits. But if it's radio waves, the wavelength of a radio wave is much bigger than that of ultraviolet. So your detector has to be bigger. Well, that's because that's why we need so many working together. Okay, right. So you have lots of them working together and they kind of work like one observatory. Got it, right? In partnership so you can make a giant array of these small satellites. Basically fake a large telescope. Yes, exactly. You fake a large telescope and you can do that in the radio relatively cheaply. Yeah. Yeah. So you know the VLA, the very large array.

We call things what they are. VLA, very large, right? It's in New Mexico. That's where they showed, that's where Jody Foster filmed the lot of her. And they all act in synchrony with the right software. They simulate a giant radio dish. The size of the grid itself. It's very cool. We got some clever people. My people are clever. Does this telescope have a serendipity mode? You mean like a fast response? No, I'm just saying. If you're only looking for what you expect to be there, you might miss the forest for the trees. Are there observations that are not targeted by some prior knowledge of what you expect? Just to see if something shows up. Well, there are survey modes

that someone, yes. The survey modes are something where you are going out looking for something, but always something surprises you. If you spend time staring at anything for long enough. With a big enough telescope. Something interesting will happen. It's hard to really spend too much time on a giant flagship mission because as we talked about earlier, there's 1,000 programs every year. Everyone wants, if you get your few little hours, I got six hours on the James Webb Space Telescope. This is very exciting and you can do amazing science. So spending, say, a month staring into space, waiting for something exciting and surprising to happen is not very easy to do on a flagship. But this is a place that the small telescopes can. And Hubble sort of did that with the Hubble Deep Field. That's right. Yeah. That's right. Yeah, the head of the Hubble. Oh, so here it was, the head of the Hubble a telescope, the director has what's called director's discretionary time, and it's handed to them to do whatever they want. For example, if there's a project, if there's a graduate student, well, it wouldn't happen in space, but if they get clouded out and they need extra time, they just hand them extra time out of that pocket, out of that repository. And he said, you know, let's just put the telescope in a blank, the blankest area of the sky, let's point the Hubble telescope to it and open the lands and just leave it there.

for like days. And people saying, you are out of your frickin' mind. And out came the Hubble Deep Field. That image of nothing was rich with a tapestry of galaxies. And that one spot on the sky, other telescopes have gone to the same place to double up on different wavelengths, different measurements. And so that's kind of what was driving my question. I don't know what the plan will be for that kind of deep field, but the director's discretionary time is something that you can use when you do have a serendipitous discovery. So we as astronomers, we run an experiment. You see something that is surprising and you don't want to wait the full year to the next deadline because you can only apply once a year. Then you put a proposal into the director and you say, director, you've got all this time.

Please, we just discovered this crazy new thing. We don't know. We need to follow it up straight away while it's still available. And then they can dole out their time based on that. This is Ken, the nerd neck Zabera from Michigan, and I support StarTalk on Patreon. This is StarTalk Radio with

Neil deGrasse Tyson. I've seen the map of the Milky Way, where they identified the 6,000 exoplanets that we've discovered. And they're hovered around our spot. in the milky way. And you zoom out, it's like, dude, we haven't explored anything. It is just all these planets are sitting on our noses, and a lot of them were discovered by, I guess, Kepler and some other space-born observatories. None of them is large as the habitable world's observatory. So how much deeper in space is that gonna take us?

Well, it depends what you mean by deeper in space. So the goals of the habitable world's observatory with respect to the search for life is really going to keep us still pretty darn close. And that is necessary because the planets we want to look at are super faint. These are rocky planets in the habitable zones of sun-like stars. And so we're still going to be limited by distance. And so we want to see the planets that are closest because they're... the brighter ones, and also because they will be separated on the sky from their central star if it's too far away. By a bigger angle. Yes, exactly. We wanted to have a bigger angle because if they're very far away, they'll be so close together that we will not be able to distinguish them. And so that is the reason to stay, say, you know, within 50 parsecs or so. Okay. And we did a whole explainer on parsecs. I'm not gonna explain that here and now.

It's a really crappy unit of distance. I apologize for it. But we're stuck with it, and Star Trek uses it, and Star Wars uses it badly. But, yeah, 50 part sex. So that's what like... Sounds really like 150 light years. Yeah, 150 something light years. Yeah, maybe we'll go beyond that. But still that little circle of all known exoplanets, it'll still be within it. Okay, and since Habitual Worlds Observatories as of this recording is yet to launch or even be built fully, right? Then your CubeSat, at some level, will be part of what's creating some foundational search data for it, is that correct? Absolutely, absolutely. So right now, the SPARC's primary mission is to study these low mass stars and understand the UV impacts on their planets. But once we are done with that, we will move on to the habitable world's observatory targets, which are...

much brighter actually than the low mass stars that will be easier for sparks to do. But that will give us a baseline characterization of the ultraviolet of

any of the planets that Hubble World Observatory will discover. And one other question, the James Webb Space Telescope, that was conceived to explore the early universe where galaxies would be born. Okay. Okay? At the time they would be born. And we know that freshly born galaxies emit a lot of ultraviolet. But they're so far away, and the light has been traveling for so long, it has redshifted from the expanding universe so that those galaxies would appear in the infrared today. OK. OK. That makes sense. You're with me. OK. So the James Webb Space Telescope was an exquisitely designed infrared telescope to snare ultraviolet light from the early universe. Okay? Yep. Turns out that's just the wavelength you want to penetrate gas clouds to see where stars are being born right in our own galaxy. So again, this was serendipity again? Well, we knew it, but that wasn't its goal because we had other infrared telescopes. This one was specific to just those galaxies, but you get to double up on its utility. So...

does the Habitable World's Observatory have another path of utility to the astrophysics community? Or is it all just for yourselves? Well, absolutely. I mean, just because the name says Habitable World's Observatory, that's only one part of the mission science goals and actually only drives one. of the instruments. We're going to have a suite of other instruments to do lots more science that are driven by completely different science. To go back to your point about the ultraviolet from galaxies, one of the driving science cases that is going to give us the requirements that we need for one or two other UV instruments is to study the UV in this nearby part of our universe, right? So at what we call low redshift, right, where these UV photons that James Webb Space Telescope is studying far into the past, we're going to be looking at more recently to study how galaxies evolve, what kind of gas is falling in, what kind of gas is being blown out, and so on. And that is a very exciting part of how it works. Okay. This magnifies, no pun intended, somebody did there, magnifies its a cross-disciplinary utility within the field. Yes, yes, it will. It is a community's telescope, right? It's not just for one group of scientists. It's gonna be for so many, a wide range of science and a wide range of instrumentation. And anyone, anyone can apply for the time when the time comes. I'll be applying. You don't need to ask me twice. But don't do it in crayons, okay?

I'm growing them now, I felt tip. So I think this might be my last question here until you come back when it launches. Okay, let's get its first data after it arrives at L2. The Hubble, which came out of a decadal survey like from the 70s, because it got built and launched, built in the 80s, launched in the 90s. The Hubble telescope was in part named after Edwin Hubble, one of his primary objectives was to establish right out of the box what the expansion rate of the universe was. Because there was uncertainty within a factor of two at the time. Okay. Yeah, and the expansion rate relates to the age of the universe. We didn't know how old the universe was to within a factor of two. It was somewhere between 10 billion and 20 billion. It was embarrassing. How accurately do we now think we have it? I'm getting there. So Hubble gets launched and so they had what was called the with flagship, there was something, a set of science goals that were prioritized. If you had other ideas, get in line, because we want these answered first. We want to know what the Hubble constant is first. And so, so much of the Hubble resources went to that. We got the Hubble constant. We got it down to like 14 billion years, plus or minus one, and it's been sharpened ever since. But notice, it was between those two estimates. Right. That's science. So we were all right. We were correct. We were correct in our uncertainty. Yes. How about that? That's a good way to say it. There you go. So are there preeminent experiments that will be conducted while everyone else has to wait because we want those known first? No, because waiting is not a very efficient use of telescope time. So we do want to survey say 150 targets for habitable planets, for sure. But we're going to have to come back and we're going to have to check them out in different parts of their orbits, and we have to revisit them. And in the time in between, we will have the complete community's creativity providing all those proposals and lots of science will be done in between. So yes, there's a prime goal to look for habitable planets, which is an incredible thing to even say out loud, right? That we live in a time that We are seriously taking this. That's an answerable question. But that is an answerable question, amazing. But in the time that, in between all that time that it's going to take, we'll be doing amazing other science, answering other equally fundamental science questions. I like the way you said that, that in the interstitial time, the creativity of the rest of the scientific community can manifest. And that's beautiful.

It is beautiful. Science is this creative activity for all of us. And those on the Decadal Survey committees, they know that whatever comes out of that, those funding priorities, if it didn't serve us all, then what are you doing? You're just bullying the funding into your own interests. Well, Evgenia, it was been a delight to have you here from Arizona. It was wonderful to be here. Thanks for coming through New York and don't let this be the last. I'll come back anytime. Thanks so much. Okay. Matt, thanks for being here. Oh, it's always a pleasure. And we're going to find you, MattKershin.com. MattKershin.com is going to my tour dates and probablyscience.com is the podcast. Probably Science, which I've been on, I think a couple of times. You've been on it. We've had you. Probably Science. It's been a treat. But I wanted you to change the name while I was on the show to be certainly Science. We'll do it for you, just for you.

All this talk of habitable worlds has got me thinking. What a luxury it is to even have those thoughts. In fact, any active generation of scientists are having thoughts that a previous generation had not even imagined were possible. And this is the hallmark of an active productive scientific research program. Be it at a university, a nation, or collectively, the world, as the world's astrophysicists tell you. And so I remember being in graduate school, people said, here's the one answer we need to know, and what is the expansion rate of the universe? And that'll make my day. And then I can die. And then we get the expansion rate of the universe, and we say, wait a minute, we are now standing on a new precipice, seeing where no one has seen before because the information and the knowledge and the wisdom and the models took us to a new place. And now I can think questions undreamt of before we arrived there. And so now I'm not asking, what's the size and age of the universe? I'm asking, is there more than one universe? I'm not asking, is there life on Mars? I'm asking, is there life on 6000 exoplanets orbiting the galaxy? So I don't even know if I can come up with a question that we might be asking ten years from now. It's frustrating, but that's a good fact. It's a good fact because, no, I can't predict where our next paths of curiosity will take us. Because as I've said before, as the area of our knowledge grows, so too does the perimeter of our ignorance. And it may be that this is an endless frontier, never ending, never coming to a stop. If it ever comes to a stop, I would be joyous and disappointed at the same time. But I'll tell you this, right now, I lose sleep at night, not wondering what the

answer is to questions I have posed. I lose sleep wondering what questions I do not even yet know. That is a cosmic perspective.

Until next time, Neil deGrasse Tyson, bidding you to keep looking.